

LAST CLASS REVIEW

What You Learned, What You Can Show, What You Can Say

Today we review the course, translate skills into interview language, and build a small GitHub artifact about one concept.

**REVIEW
+
PORTFOLIO**

Final project due May 26, 2026

TODAY'S PATH

The class ends by turning learning into evidence

- Review the course in plain language
- Practice interview and resume language
- Walk through the Week 15 notebooks
- Create one GitHub concept artifact
- Leave with a final project next step

COURSE IN ONE SENTENCE

Security is not just blocking hackers; it is protecting data responsibly

**Protect data
across its
lifecycle.**

- Who can access it?
- How is it stored or shared?
- How is it analyzed?
- Who could be harmed?
- What evidence supports the decision?

FOUNDATION

The CIA triad is a simple way to organize security goals

Confidentiality

Only the right people can access the data.

Integrity

The data stays accurate, complete, and trustworthy.

Availability

The system or data is usable when it is needed.

RISK THINKING

A beginner security analyst does not treat every problem as equal

- Risk asks what could go wrong
- Impact asks how serious the harm could be
- Likelihood asks how plausible the problem is
- Evidence asks what we actually know
- Priority asks what should be handled first

IDENTITY

Authentication and access control answer different questions

Authentication

Who are you? Example: login, password, MFA.

Authorization

What are you allowed to do? Example: read, edit, export.

Least privilege

Give only the access needed for the task.

ENCRYPTION

Cryptography protects data, but only when the system around it is designed well

- Encryption can protect data at rest or in transit
- TLS helps protect web traffic in transit
- Hashing is useful for password storage discussions
- Keys, configuration, and access still matter
- Encryption does not solve every privacy problem

SYSTEMS

Web, network, database, and cloud security are connected

- Web apps handle input, sessions, and user actions
- Networks move traffic between systems
- Databases store structured records
- Cloud platforms add sharing, permissions, and vendor responsibility
- Weakness in one layer can affect the others

DATA REPRESENTATION

Tables, CSVs, JSON, and documents shape what analysis can show

CSV / table

Rows and columns make comparison and sorting easy.

JSON / document

Nested structures can represent flexible real-world objects.

Data dictionary

Definitions make analysis safer and clearer.

PRIVACY AND ETHICS

Data can harm people even when it looks technical

- Identifiers can be direct or indirect
- Categories can shape how people are seen
- Missing data can create unfair conclusions
- Surveillance can be excessive even when technically possible
- Ethics asks who benefits, who is burdened, and who gets a choice

SECURITY ANALYTICS

Logs and metrics are evidence, not automatic truth

- A log shows recorded activity, not the whole story
- A metric summarizes signals, but it can hide context
- A chart can look convincing even when data is incomplete
- Good analysts explain both findings and limits
- Recommendations should match the evidence

ADVERSARIAL THINKING

Data poisoning and evasion teach us to question inputs and outputs

Data poisoning

Bad or manipulated data can shape bad outputs.

Evasion

A system can be fooled during use.

Limitation

A careful recommendation says what it cannot prove.

INCIDENT RESPONSE

After something goes wrong, calm evidence beats panic

- Separate confirmed facts from assumptions
- Contain the immediate harm
- Preserve logs and evidence
- Identify what data was involved
- Communicate clearly and avoid unsupported claims

CAREER TRANSLATION

Do not say “I learned cybersecurity.” Say what you can actually do.

**Specific beats
vague.**

- I can explain access control with examples
- I can document dataset fields and limits
- I can read simple security metrics
- I can separate facts from assumptions
- I can write a realistic recommendation

INTERVIEW FORMULA

Use concept, evidence, workplace value, and limit

1

Concept: name the idea

2

Evidence: name the lab, dataset, source, or project

3

Value: explain why it matters at work

4

Limit: say what you would check before a stronger claim

GITHUB PORTFOLIO

A small clear artifact is better than a big confusing one

- One concept from the course
- Beginner-friendly explanation
- Simple example or code snippet
- Privacy or ethics note
- Workplace/interview sentence
- Limitation section

NOTEBOOK 1

The course review notebook maps concepts to language

- Review major course concepts
- Search by keyword
- Print a concept card
- Generate resume bullet starters
- Export a small planning CSV

NOTEBOOK 2

The GitHub starter notebook generates a README draft

- Students edit concept fields
- Run a tiny optional code example
- Generate Markdown text
- Save a README file
- Upload or revise for GitHub

FINAL LAB

Create one GitHub artifact that teaches one concept

- Choose a concept
- Create a README or notebook
- Explain why the concept matters
- Add an example, table, checklist, or code
- Write one interview sentence
- Submit a link or draft by the end of class

EXIT AND FINAL PROJECT

Leave with one artifact, one sentence, and one next step

Due May 26, 2026

- Artifact: concept guide or code example
- Sentence: what you learned and can explain
- Next step: what remains for the final project
- Reminder: no presentation